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13. Sodium sesquicarbonate is a white crystalline solid which is used in bath salts, water treatment and as a cleaning and laundry product. Its formula can be represented as follows.

$$Na_2CO_3.\boldsymbol{x}NaHCO_3.\boldsymbol{y}H_2O$$

A student is asked to carry out an experiment to find the value of $\boldsymbol{x}$ using the following double titration method.

- Weigh out accurately about 1.7 g of sodium sesquicarbonate using a two-decimal place balance. Dissolve and make up to 250 cm^3 in a volumetric flask in the usual way.
- Measure 25.0 cm^3 of this solution into a conical flask and add a small amount of phenolphthalein indicator.
- Add 0.100 mol dm^{-3} hydrochloric acid from a burette whilst swirling the mixture until the phenolphthalein changes colour from **pink to colourless**.
- Record the results.
- Add a few drops of methyl orange indicator and continue titrating until the methyl orange changes colour from **yellow to orange**.
- Record the results.

The results of the titrations are shown below.

		1	2	3	4
With phenolphthalein (**A**)	Final volume / cm^3	7.80	30.70	7.75	30.25
	Initial volume / cm^3	0.00	23.30	0.30	22.75
	Titre **A** / cm^3	7.80	....	....	....
With methyl orange (**B**)	Final volume / cm^3	23.30	45.60	22.75	45.10
	Initial volume / cm^3	7.80	30.70	7.75	30.25
	Titre **B** / cm^3	15.50	....	....	....

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(a) Complete the table and calculate the mean titre at the phenolphthalein end-point (**A**) and the mean titre at the methyl orange end-point (**B**). [3]

Mean titre **A** = ... cm^3

Mean titre **B** = ... cm^3

(b) Name the piece of apparatus the student should have used to transfer 25.0 cm^3 of the solution into the conical flask. [1]

....

(c) The percentage error due to the burette in the first titration with phenolphthalein is just over 1%. Suggest what the student could have changed in the method to halve this figure. [1]

....

....

(d) Despite having a larger percentage error due to the burette, suggest a reason why mean titre **A** could be more accurate than mean titre **B**. [1]

....

....

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(e) (i) The equation for the reaction taking place in the phenolphthalein titration is as follows.

$$CO_3^{2-} \quad + \quad H^+ \longrightarrow HCO_3^-$$

Mean titre **A** is the volume needed to convert all the carbonate ions present to hydrogencarbonate ions.

Use this information and the student's results to calculate the number of moles of carbonate present in the original solution. [1]

$n(Na_2CO_3)$ = ... mol

(ii) The equation for the reaction taking place in the methyl orange titration is as follows.

$$HCO_3^- \quad + \quad H^+ \longrightarrow CO_2 \quad + \quad H_2O$$

Mean titre **B** is the volume needed to convert all the hydrogencarbonate ions present to carbon dioxide and water.

This hydrogencarbonate is from the original hydrogencarbonate **and** the converted carbonate, therefore

moles HCl in mean titre **B** = moles $NaHCO_3$ + moles Na_2CO_3

Use this information and the student's results to calculate the number of moles of hydrogencarbonate present in the original solution. [2]

$n(NaHCO_3)$ = ... mol

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(iii) Use your answers to parts (i) and (ii) to show that the value of x in $Na_2CO_3.xNaHCO_3.yH_2O$ is 1. [1]

(iv) The relative formula mass of $Na_2CO_3.xNaHCO_3.yH_2O$ is 226. Calculate the value of y. [2]

y = ...

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END OF PAPER